

Snap-Load Transmission Between Cables Sharing a Towed Payload: Why the Impulsive Estimate Overpredicts

Hadi Hajieghrary¹ and Paul Schmitt²

Abstract—When several robots or vessels tow a shared payload with cables, a re-engaging slack cable produces a snap load that can unload the neighboring cables. A natural rigid-impact (impulsive) estimate treats the re-engagement as instantaneous and holds the neighbors’ far ends fixed. For a collinear tow of unilateral Kelvin–Voigt cables, we show that these assumptions require opposing mass regimes: brief contact requires light towing vehicles, whereas fixed far ends require heavy ones. We derive a similarity law under which the transmitted fraction of the snap peak is independent of closing speed and, without dissipation, of cable stiffness; an exact unloading criterion; and a closed-form law under a half-sine contact pulse which, with free far ends, lies at least 21% below the corresponding impulsive value for up to ten cables. Nonlinear planar simulations of five controlled tugs in sixteen configurations test a forecast of 16.5%, committed before any neighbor-tension drop was computed, against the rigid-impact 44%. The pre-registered statistic stays well below the impulsive range in every configuration and rises with stiffness as predicted (by 3.6 against 4.5 percentage points), whereas a pre-registered pairwise test and a later fixed-dissipation control fail. A declared test at a second mass ratio supports the predicted rise (6.5 against 7.0 percentage points). Post hoc counterfactual re-integrations give a median transmission of 14.8%; subtracting a time-shifted background drop instead gives a lower bound of 9.3%. The rigid-impact baseline thus overstates cross-cable transmission roughly threefold.

I. INTRODUCTION

Cable-connected teams let several robots or vessels move payloads that one vehicle could not, as in cooperative towing [1]–[3], multi-UAV transport [4], [5], and cable-driven robots [6]. Cables pull only while taut: a cable unloaded by payload oscillation, maneuvers, or disturbances snaps when pulled taut again, and because the cables share the payload, the snap changes every neighbor’s tension; a neighbor that loses enough may go slack, letting slack spread.

We ask: *given the peak load in a snapping cable, how much tension can a neighboring cable lose?* The answer is the cross-cable snap-transmission ratio, the largest reduction in a neighbor’s tension divided by the snapping cable’s peak. It links a fast transient to the tension margin that governs pretension selection, tension allocation, and slack screening. The natural estimate treats the re-engagement as an instantaneous impulse and holds the neighbors’ far ends fixed. Yet the snap’s duration and the neighbors’ motion are set by the same stiffnesses and masses, so a consistent model must retain both finite contact and endpoint recoil. We take

the snap peak as an input; predicting the snap itself, or a universal design bound, is not our aim.

Our contributions are the following.

- 1) **Formulation and structural diagnosis.** We define the transmission ratio and, in the collinear model, an exact criterion for complete unloading of a neighbor. We show that the rigid-impact baseline needs light vehicles for an impulsive contact but heavy ones for held far ends; with free far ends and identical vehicles, the neighbor mode is never slower than the isolated-pair contact, so the impulsive regime is unreachable at every mass ratio.
- 2) **Finite-duration transmission theory.** For a collinear tow of unilateral Kelvin–Voigt cables, a similarity law makes the ratio independent of closing speed and, without dissipation, of stiffness. Under a half-sine contact pulse the ratio is closed-form in mass ratio and cable count and, with free far ends, at least 21% below the corresponding impulsive value for up to ten cables. Self-consistent contact is treated numerically.
- 3) **Evaluation in a nonlinear controlled fleet.** A planar simulation of five controlled tugs in sixteen configurations is scored against a forecast committed before any neighbor-tension drop was computed, and a later declared test adds a second vessel mass. The evidence, summarized in Table III, supports the predicted population-level magnitude and parameter dependence, not a pairwise distribution or a universal bound.

II. RELATED WORK

Nonsmooth mechanics represents a newly active unilateral constraint by an impulse and a velocity jump; the Delassus operator, the inverse inertia seen by the constraints, maps constraint impulses to velocity changes [7], [8], and appears in robotics as the inverse operational-space inertia [9]. Compliant-impact theory shows that the outcome can instead depend on relative stiffnesses and stored energy [10], and impact-aware planning treats the slack–taut transition of a cable-suspended aerial robot as a hybrid impact event [11]. None of these determines the peak unloading of one cable when another re-engages through a shared movable payload; we use the rigid-impact construction only as a baseline.

Marine snap-loading work focuses on the force *within the snapping line*: the dimensionless snap susceptibility of a package suspended from multiple cables [12], lumped slack–taut cable dynamics [13], continuous cables coupled through a submerged body and deck machinery [14], first-snap forces

¹Hadi Hajieghrary (e-mail: hadi.hajieghrary@gatech.edu).

²Paul Schmitt (e-mail: pauls@massrobotics.org, paul.schmitt@reynolds-moore.com).

The authors prepared this manuscript and present it solely in their individual capacities.

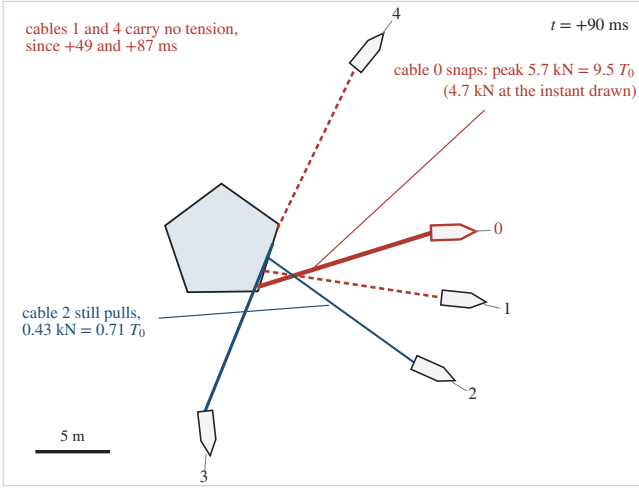


Fig. 1. A selected, atypical snap on the testbed ($T_0 = 0.6$ kN), 90 ms after cable 0 re-engages; taut cables solid (width growing with tension), slack cables dashed. Cable 0 peaks at $9.51T_0$, far above the scored range $[T_0, 4T_0]$, at 58 ms. Cable 1, already near slack ($0.66T_0$), reaches zero at 49 ms; cables 4 and 3 follow at 87 and 206 ms; cable 2 stays taut for one engagement period.

in synthetic ropes [15], extreme mooring tensions in slack-taut events [16]–[18], and finite-element and discontinuous-Galerkin models that resolve axial waves [19]. These models are far richer than our lumped one, but none, including the multi-cable formulation of [14], defines the normalized peak transfer from a snapping cable to a neighbor or asks how freely moving endpoints alter it.

Cooperative transport research models several cables on a movable payload to address equilibrium, feasible tensions, trajectories, and manipulation [4], [20], or towing-force allocation under disturbances [3], to keep transport feasible rather than to quantify the transient transmitted between cables.

III. MODEL AND PROBLEM

A. The Fleet and the Question

A fleet of N vessels tows one payload, each through one cable loaded to the pretension T_0 in steady tow. The cables are unilateral Kelvin–Voigt lines, with tension $[ke_i + c\dot{e}_i]_+$ for elongation $e_i > 0$ and none otherwise [21]; the clamp keeps the damper from pushing in compression just before separation [22]. When cable j re-engages with closing speed $V = \dot{e}_j(0^+) > 0$, its tension reaches T_p and returns to zero after a finite contact. We call this a *snap* (Fig. 1) and define the transmission to neighbor i by

$$\rho_{ij} = \frac{\max_t \Delta T_i(t)}{T_p}, \quad T_p = \max_t T_j(t), \quad (1)$$

where ΔT_i is the drop relative to the counterfactual tension without the snap. Section V-B realizes the counterfactual numerically.

B. Collinear Reduction

Section V measures the cost of the following idealization on a planar fleet.

Assumption 1. The cables are parallel and attached so a snap does not rotate the payload; all have stiffness k and damping c , the vessels share mass m_A and drag c_A , and the payload has mass m_L and drag c_L . During one snap, thrust and weather forces are constant, so they cancel in the perturbation about the steady tow, and the $N - 1$ neighbors begin from a common taut state of tension $T_n > 0$.

Let x , y_j , and y be payload, parent-vessel, and neighbor-vessel displacements from steady tow; by symmetry and uniqueness the neighbors move together.

Assumption 2. Solutions cross the switching surface $e_j = 0$ transversally, so the hybrid solution is unique.

While the neighbors remain taut,

$$\begin{aligned} m_L \ddot{x} &= f + (N - 1)\delta T - c_L \dot{x}, \\ m_A \ddot{y} &= -\delta T - c_A \dot{y}, \\ m_A \ddot{y}_j &= -f - c_A \dot{y}_j, \end{aligned} \quad (2)$$

with $\delta T = k(y - x) + c(\dot{y} - \dot{x})$, $f = [ke_j + c\dot{e}_j]_+$ for $e_j = y_j - x > 0$ and $f = 0$ otherwise, and zero initial state except $\dot{y}_j(0) = V$. Hence $\Delta T = -\delta T$ and $\rho = \max \Delta T / \max f$ over the first contact and its response; a later contact of cable j is a new snap. For prescribed f , (2) is linear, so the snap contribution is a superposed perturbation; the nonlinear planar testbed instead uses paired re-integrations with and without the snapping-cable force, a counterfactual rather than a superposition.

C. The Impulsive Estimate

Replace the contact by an impulse $J = \int f dt$, hold the neighbor vessels ($y \equiv 0$), and ignore dissipation. The payload leaves with velocity J/m_L and oscillates at $\omega_L = \sqrt{(N - 1)k/m_L}$, giving $\max \Delta T = kJ/(m_L \omega_L)$. For an isolated half-sine contact $f = T_p \sin \omega_e t$ on $[0, \pi/\omega_e]$, with $\omega_e = \sqrt{k(1/m_A + 1/m_L)}$, $J = 2T_p/\omega_e$ and

$$\rho_{\text{imp}} = \frac{2k}{m_L \omega_L \omega_e}. \quad (3)$$

Equivalently, $\rho_{\text{imp}} = (2k/(\omega_L \omega_e)) [W^\top M_L^{-1} W]_{ij}$, where W has the cable wrenches $(d_i, r_i \times d_i)$ as columns, with unit chord d_i and attachment arm r_i , and $M_L = \text{diag}(m_L, m_L, I_L)$; under Assumption 1 the lever arms vanish and the off-diagonal entry is $1/m_L$. At $N = 5$ and $m_A/m_L = 0.24$, (3) gives $\rho_{\text{imp}} = 0.440$, so a neighbor at T_0 would unload near $2.3T_0$. It is the direct impact construction, not a published law; we use it as the baseline and remove its finite-contact and held-end assumptions below.

IV. TRANSMISSION THEORY

A. Similarity and Threshold Laws

Proposition 1. Let Assumptions 1 and 2 hold and let the neighbors remain taut. Define $\gamma = c/\sqrt{km_L}$, $\gamma_L = c_L/\sqrt{km_L}$ and $\gamma_A = c_A/\sqrt{km_L}$. Every tension in (2) equals $V\sqrt{km_L}$ times a function of $t\sqrt{k/m_L}$ that depends only on m_A/m_L , N , γ , γ_L and γ_A . Consequently $\rho = \Phi(m_A/m_L, N, \gamma, \gamma_L, \gamma_A)$ for a function Φ that does not

depend on V , k or T_n separately, and without dissipation ρ depends on m_A/m_L and N alone.

Proof. Set $\ell = V\sqrt{m_L/k}$ and $s = t\sqrt{k/m_L}$, with $x = \ell\xi$, $y = \ell\eta$, and $y_j = \ell\eta_j$. Inertial, spring, and damping terms in (2) share the force scale $k\ell = V\sqrt{km_L}$; after division by it, the equations and switching surface depend only on $r = m_A/m_L$, N , γ , γ_L , and γ_A , with $\eta'_j(0) = 1$. Uniqueness fixes the corresponding dimensionless response. Since both f and δT scale by $k\ell$, their peak ratio has the stated dependence, and T_n does not enter (2). $\square$

Without solving (2), the proposition predicts that stiffness can change the transmission only through the three dissipation numbers, which fall as $k^{-1/2}$; an undamped law such as (3) cannot depend on stiffness at all (tested in Section V-C). The same scaling gives an exact criterion for slackening a neighbor, that is, for its tension to reach zero. Let $\bar{T}_p$ denote the peak that the taut model (2) assigns to a given closing speed; it equals the actual peak whenever the neighbors stay taut.

Corollary 1. *Under Assumptions 1 and 2, a snap slackens the neighbors if and only if $\rho\bar{T}_p \geq T_n$.*

Proof. The minimum taut-model tension is $T_n - \rho\bar{T}_p$. If positive, uniqueness keeps the actual solution taut. Otherwise, the actual and taut solutions coincide until the first zero of $T_n + \delta T$, so the neighbor reaches zero tension. $\square$

Control implication. Since T_n does not enter (2), the proof extends to unequal pretensions $T_{n,i}$: no neighbor slackens if and only if $\rho\bar{T}_p < \min_i T_{n,i}$. The bound is joint, since once a weaker neighbor slackens the others unload further, so a pretension or allocation policy must enforce it for all neighbors at once, with $\bar{T}_p \propto V$ predicted from the anticipated closing speed (Proposition 1) and a margin for model error.

B. Closed Form Under the Contact Pulse

To obtain ρ in closed form, we replace the self-consistent contact by the isolated-pair pulse of (3) and set aside dissipation; Sections IV-C and IV-D restore both. The ratio of the peak half-sine response of a linear oscillator to its impulsive response is the pulse's shock spectrum [23]; we need it with an argument that identifies which extremum is the peak.

Lemma 1. *Let $0 < \varepsilon \leq 3$ and let u solve $\ddot{u} + \varepsilon^2 u = \sin t$ on $[0, \pi]$ and $\ddot{u} + \varepsilon^2 u = 0$ afterwards, with $u(0) = \dot{u}(0) = 0$. Then $\sup_{t \geq 0} |u(t)| = 2S(\varepsilon)/\varepsilon$, where*

$$S(\varepsilon) = \begin{cases} \frac{\cos(\pi\varepsilon/2)}{1 - \varepsilon^2}, & \varepsilon < 1, \\ \pi/4, & \varepsilon = 1, \\ \frac{\sin(2\pi/(1 + \varepsilon))}{2(\varepsilon - 1)}, & 1 < \varepsilon \leq 3. \end{cases} \quad (4)$$

Moreover $S(\varepsilon) \rightarrow 1$ as $\varepsilon \rightarrow 0$, and S decreases on $[1, 3]$ from $\pi/4$ to $1/4$.

Since the same impulse, $\int_0^\pi \sin t \, dt = 2$, applied at $t = 0$ gives amplitude $2/\varepsilon$, S is the peak relative to the impulsive one.

Proof. For $\varepsilon \neq 1$,

$$u = \frac{\sin \varepsilon t - \varepsilon \sin t}{\varepsilon(1 - \varepsilon^2)}, \quad \dot{u} = \frac{2 \sin \frac{(1+\varepsilon)t}{2} \sin \frac{(1-\varepsilon)t}{2}}{1 - \varepsilon^2}.$$

The energy $E = u^2 + \dot{u}^2/\varepsilon^2$ satisfies $\dot{E} = 2\dot{u} \sin t/\varepsilon^2$ on $[0, \pi]$, is constant afterwards, and bounds $|u| \leq \sqrt{E}$. For $\varepsilon < 1$, both sine factors are positive on $(0, \pi]$, so $\dot{u} > 0$, E is nondecreasing, and the supremum is $\sqrt{E(\pi)}$. For $1 < \varepsilon \leq 3$, $\sin((\varepsilon - 1)t/2) > 0$ on $(0, \pi)$, while $\sin((1 + \varepsilon)t/2)$ is positive before $t^* = 2\pi/(1 + \varepsilon)$ and, because $(1 + \varepsilon)\pi/2 \leq 2\pi$, nonpositive on $[t^*, \pi]$; hence u increases to t^* , E is nonincreasing afterwards, and $|u(t)| \leq \sqrt{E(t^*)} = u(t^*) = \sin t^*/(\varepsilon(\varepsilon - 1))$ for $t \geq t^*$, using $\varepsilon t^* = 2\pi - t^*$. At $\varepsilon = 1$, $u = (\sin t - t \cos t)/2$ with $\dot{u} = (t \sin t)/2 \geq 0$, so the supremum is $\sqrt{E(\pi)} = \pi/2$. Dividing by $2/\varepsilon$ gives (4). On $(1, 3]$, $w = \pi - t^* \in (0, \pi/2]$ increases with ε and $S = (\pi - w) \sin w/(4w)$, whose w -derivative has the sign of $w(\pi - w) \cos w - \pi \sin w < 0$, since $\pi \sin w \geq \pi w \cos w > (\pi - w)w \cos w$ for $\cos w > 0$ and the first term vanishes at $w = \pi/2$. $\square$

Theorem 1. *Let Assumption 1 hold with $c = c_L = c_A = 0$, and replace the contact force in (2) by $f(t) = T_p \sin \omega_e t$ on $[0, \pi/\omega_e]$ and $f = 0$ afterwards. Let $\omega_q = \omega_L$ if the neighbor vessels are held and $\omega_q = \omega_f := \sqrt{k((N - 1)/m_L + 1/m_A)}$ if they are free. If $\varepsilon_q := \omega_q/\omega_e \leq 3$, then*

$$\rho = \frac{2k}{m_L \omega_q \omega_e} S(\varepsilon_q). \quad (5)$$

Proof. For $q = x - y$, (2) gives $\ddot{q} + \omega_q^2 q = f/m_L$ (with $q = x$ when held) and $\Delta T = kq$. Setting $s = \omega_e t$ and $q = T_p u(s)/(m_L \omega_e^2)$ reduces the response to Lemma 1, yielding (5). $\square$

The prefactor in (5) is the impulsive transmission of the same fleet, and (3) is the held case with $S = 1$. With free far ends and $N \leq 10$, the theorem holds for every choice of masses:

Corollary 2. *With free far ends and the half-sine pulse of Theorem 1,*

$$\frac{2k}{m_L \omega_f \omega_e} = \frac{2m_A}{\sqrt{(m_A + m_L)(m_L + (N - 1)m_A)}}, \quad \varepsilon_f^2 = 1 + \frac{(N - 2)m_A}{m_A + m_L}. \quad (6)$$

Hence for $2 \leq N \leq 10$ and all masses $\varepsilon_f \geq 1$, with $\varepsilon_f = 1$ when $N = 2$ and $1 < \varepsilon_f < \sqrt{N - 1} \leq 3$ when $N \geq 3$, and that pulse removes at least $1 - \pi/4 \approx 21\%$ of the corresponding impulsive transmission $2k/(m_L \omega_f \omega_e)$, with equality only for $N = 2$; since $\omega_f > \omega_L$, the free-end law (5) lies more than $1 - \pi/4$ below (3).

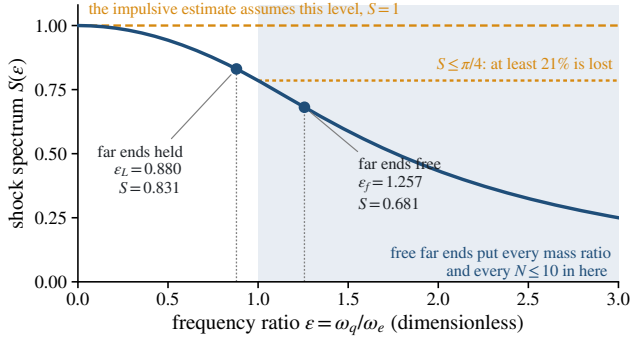


Fig. 2. Half-sine shock spectrum $S(\varepsilon)$; the peak response to the pulse relative to the same impulse delivered instantly; (3) assumes $S = 1$. Held far ends put the testbed at $\varepsilon_L = 0.880$, $S = 0.831$; free far ends force $\varepsilon_f \geq 1$, so $S \leq \pi/4$ for $N \leq 10$, and put the testbed at $\varepsilon_f = 1.257$, $S = 0.681$.

Proof. Substitution gives (6). Thus $\varepsilon_f = 1$ for $N = 2$ and $1 < \varepsilon_f < \sqrt{N-1} \leq 3$ for $3 \leq N \leq 10$; Lemma 1 then gives $S(\varepsilon_f) \leq S(1) = \pi/4$. $\square$

The impulsive regime needs $\varepsilon_q \rightarrow 0$, but with free far ends the neighbor mode is never slower than the pulse (Fig. 2). The held case shows why (3) cannot be rescued by choosing the masses: $\varepsilon_L^2 = (N-1)m_A/(m_A + m_L)$ is small, and the contact nearly impulsive, only when $m_A \ll m_L/(N-1)$, whereas held far ends require $\omega_f \approx \omega_L$, that is $m_A \gg m_L/(N-1)$. The two hypotheses demand opposite mass ratios, and the testbed, with $m_A/m_L = 0.24$ against $1/(N-1) = 0.25$, satisfies neither. Removing them one at a time, the finite pulse costs 17% of (3); the recoil of the far ends removes a further 43% of what remains, and damping and drag 21% of the rest, while the pulse shape assumed in Theorem 1 changes the result by 1–2%.

C. Self-Consistent Contact Across Mass Ratios

We integrate (2) with its unilateral force (adaptive RK4(5), relative tolerance 10^{-10} , maximum step $\pi/(800\omega_e)$); with the prescribed pulse instead, this reproduces (5) to six digits. For five cables the half-sine law is within 1.7% of the undamped self-consistent ratio for $m_A/m_L \leq 1$; the latter rises with vessel mass while far-end recoil dominates and falls once the lengthening contact does, with a secondary rise beyond $m_A/m_L \approx 5$ (Fig. 3). We evaluate it on a 31-point logarithmic grid of m_A/m_L over $[10^{-1.5}, 10^{1.5}]$ and refine the grid maximum by a bounded Brent search to 10^{-3} . For $N = 5$ this gives $\rho^* = 0.367$ at $m_A/m_L = 1.41$, a numerical maximum, not a proof; by Corollary 1, no snap with $\bar{T}_p < T_n/\rho^* \approx 2.73T_n$ slackens a neighbor in the undamped collinear model. For $N = 3$ –8 (Table I), every maximum lies more than a decade inside the interval and $(N-1)\rho^*$ decreases monotonically from 1.54 to 1.42; whether this reflects a bound $\rho^* \lesssim C/(N-1)$ is open, and for $N = 2$ the search gives $\rho^* = 1.41$ at $m_A/m_L = 4.0$.

D. Dissipation and Stiffness

With the testbed’s cable damping and drag (Table II), the self-consistent ratio is 0.167 and the half-sine ratio 0.165; the

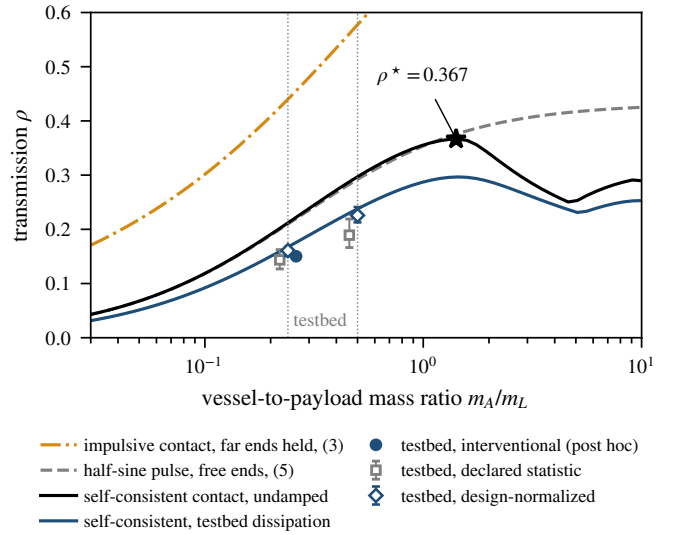


Fig. 3. Five-cable transmission versus mass ratio. The undamped self-consistent contact has a numerical interior maximum $\rho^* = 0.367$ (star); the half-sine law tracks it for $m_A/m_L \leq 1$. Testbed markers, for the nominal cell of Section V and its paired cell at $m_A = 1250$ kg: the recorded-geometry statistic (squares) and the design-normalized statistic (diamonds), both with seed intervals, the latter post hoc at 0.24 and the declared primary of the second-mass test at 0.50; and, at 0.24, the post hoc interventional swing (circle).

TABLE I
LARGEST UNDAMPED SELF-CONSISTENT TRANSMISSION IN THE
MASS-RATIO SEARCH

N	m_A/m_L at max	ρ^*	$1/\rho^*$	$(N-1)\rho^*$
3	2.67	0.770	1.30	1.54
4	1.94	0.501	2.00	1.50
5	1.41	0.367	2.73	1.47
6	1.11	0.289	3.46	1.45
7	0.92	0.239	4.19	1.43
8	0.78	0.203	4.92	1.42

latter coincides to six digits with the planar forecast committed for the test of Section V. By Proposition 1 the stiffness enters only through the dissipation numbers, which yields two predictions. First, quartering and quadrupling k moves ρ from 0.139 to 0.184 under the half-sine pulse and from 0.140 to 0.186 self-consistently, increases of 0.045 and 0.046, whereas every undamped law, (3) and (5) included, predicts none. Second, the swing’s extremum, which Lemma 1 places at $2\pi/((1+\varepsilon_f)\omega_e) = 149$ ms without dissipation and which then scales as $k^{-1/2}$, arrives earlier with dissipation and scales more weakly: at $k/4$, k and $4k$ the model places it at 205, 125 and 69 ms self-consistently and at 250, 135 and 71 ms under the half-sine pulse. Only the half-sine values of the first prediction were committed before the campaign, as the planar forecast; the self-consistent and timing values were computed after the measurements of Section V-C.

V. SIMULATION OF A MULTIBODY FLEET

We test the collinear predictions on a planar Drake model [24] of five tugs and one payload (Table II). The payload is a regular pentagon of circumradius 4 m with bow attachments

TABLE II
TESTBED PARAMETERS

Quantity	Symbol	Value
Payload mass, drag	m_L, c_L	2500 kg, 5500 N s/m
Vessel mass, drag	m_A, c_A	600 kg, 350 N s/m
Cable stiffness, damping	k, c	1.7×10^5 N/m, 1800 N s/m
Cable rest length	L	12 m
Payload, vessel yaw inertia	I_L, I_A	2.0×10^4 , 500 kg m ²
Payload, vessel yaw drag		2.0×10^4 , 800 N s
Heading gain (nominal, doubled)		500, 1000 N m/rad
Cable-bearing trim gain		100 N m
Steady thrust (parallel cells)	F_T	$1.32 T_0$
Weather std, payload/vessel	σ_W	3.5, 0.7 kN
Weather memory	τ_W	8 s
Engagement, payload modes	ω_e, ω_L	18.7, 16.5 rad/s

spanning $\pm 31.7^\circ$, and each vessel attaches 1.5 m astern of its center; parallel cells keep the nominal chords parallel, fan cells spread them over ± 0.55 rad. Each vessel holds its steady-tow thrust open-loop and runs a 50 Hz heading loop, proportional in heading error with a cable-bearing trim term; every body receives first-order autoregressive weather with 8 s memory. The plant is integrated at 0.5 ms. An isolated vessel–payload refinement passes a 0.5% energy-envelope criterion at 0.5 ms with little margin (0.47%; 0.95 and 0.23% at 1.0 and 0.25 ms) and so selected 0.25 ms; the campaign uses 0.5 ms for cost, at which the five-vessel engagement peak lies within 0.08% of its 0.25 ms value. As the cable force is explicit, the error scales with $\omega_e h$ for step h , so the $4k$ cells sit at the nominal 1.0 ms level. A check declared before its runs re-ran the nominal and $4k$ cells on all 22 seeds at 0.25 ms, re-integration of Section V-B included: the declared and design-normalized statistics and the interventional swing moved by at most 0.006 and its lag by under 1 ms, each change inside its paired interval and below the half-width of its seed interval.

The campaign has sixteen cells: ten parallel cells spanning $T_0 \in \{0.6, 1.0, 1.4, 1.8\}$ kN, weather intensity 0.5 or 1 (Table II lists intensity 1), and nominal or doubled heading gain; two stiffness cells at $k/4$ and $4k$; and four fan cells. Each cell runs the same 22 seeds, of which 20 are scored; the two pilot seeds, reserved by the declaration for fitting, fit only the neighbor-margin law of the slack-onset test cited in Section VI. Re-engagements of one cable within 2 s of its first form one event, and a scored observation pairs that first re-engagement, with $T_p \in [T_0, 4T_0]$ and all other cables taut, with one neighbor. Drops are read over one engagement period $P = 2\pi\sqrt{\mu/k}$, $\mu = m_A m_L / (m_A + m_L) = 484$ kg. Intervals are 95% bootstraps over whole seeds, with 2000 replicates unless a caption states otherwise.

A. Forecast, Protocol, and Evidence Status

We committed the forecast and every pass criterion to a hashed declaration [25], [26] before launching the campaign and before any neighbor-tension drop had been computed. Ten of the sixteen cells (220 of 352 runs) re-simulate, bit for bit, an earlier campaign on the same seeds, whose stored records (slack-event times and peaks, 1 Hz tension samples)

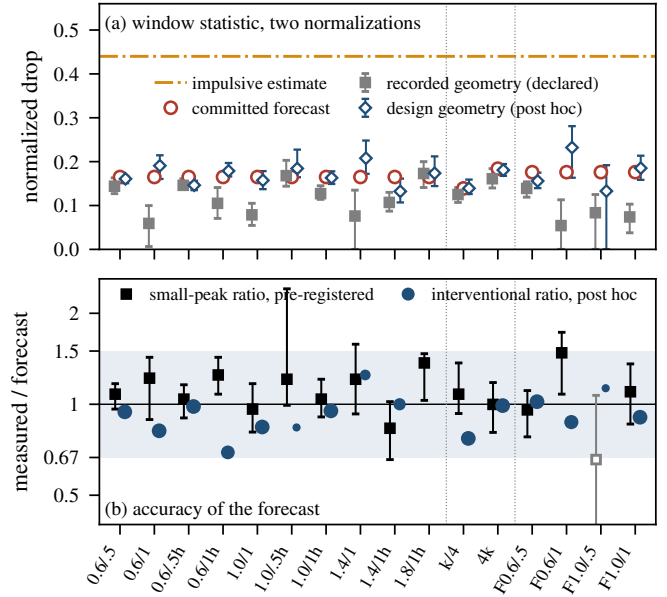


Fig. 4. Transmission in sixteen cells, labeled by T_0 in kN and weather intensity (h: doubled heading gain; F: fan; $k/4$, $4k$: stiffness cells). (a) Declared recorded-geometry statistic (squares) and post hoc design-geometry reading (diamonds), with seed intervals, against the committed forecast (circles). (b) Measured over forecast: pre-registered small-peak ratio (squares, seed intervals; open: unscored cell) and post hoc interventional ratio (circles, area growing with observation count); the band is the declared factor 1.5.

cannot yield the declared statistic; the forecast has no fitted input. The forecast R_{ij} is the peak tension reduction in cable i per unit half-sine peak applied through cable j , from a 34-state translation-invariant linearization of the closed-loop planar fleet about steady tow that retains drag, cable damping, and heading control; the snapping cable contributes only its pulse wrench, and the outputs are $k\delta e_i + c\delta \dot{e}_i$. Central-difference Jacobians use a step of 10^{-5} per reduced coordinate; steps of 10^{-4} and 10^{-6} change every R_{ij} by less than $10^{-4}\%$.

The declared (pre-registered) window statistic divides the measured tension reduction by $T_p g_{ij}$, where g_{ij} is the (i, j) entry of $G = (2k/(\omega_L \omega_e))W^\top M_L^{-1}W$ (Section III) divided by (3). At the design geometry of the parallel cells the off-diagonal g_{ij} range from 0.50 to 1.23; the statistic instead evaluates g_{ij} at the recorded geometry of each event, where it can approach or cross zero ($g_{ij} \leq 0$ for 14–39% of a cell's scored observations). The cell forecast is the median of R_{ij}/g_{ij} over the 20 ordered pairs at the design geometry: 0.165 in the ten parallel cells at nominal stiffness, 0.139 and 0.184 at the stiffness extremes, and 0.176 in the fan cells. Table III gives the status of each analysis, that is, when it was specified relative to the data it evaluates.

B. Magnitude

The declared statistic stays below the impulsive band $[0.2, 0.9]$ in every cell (Fig. 4a) but below the forecast in 14 of 16. On the 3008 re-integrated observations introduced below, an additive background floor ($0.057T_p$ in time-shifted windows against $0.154T_p$ at snaps) would raise it; the offset

TABLE III
EVIDENCE STATUS AND PRINCIPAL OUTCOMES

Analysis	Status	Prediction or benchmark	Observed outcome	Interpretation
Impulsive band	Pre-registered	No cell in $[0.2, 0.9]$	None enters; cell medians 0.054–0.173	Rejects the rigid-impact magnitude
Small-peak accuracy	Pre-registered	Ratio to forecast within $\times 1.5$, $T_p < 2.3T_0$	0.83–1.48 in all 15 evaluable cells	Forecast within declared factor
Stiffness trend	Pre-registered	$+0.045$ from $k/4$ to $4k$	$+0.036$ [$+0.015, +0.058$]	Supports dissipative dependence
Pairwise ranking	Pre-registered	Pair order of R_{ij}	Met in 2 of 15 evaluable cells	No pairwise claim
Fixed dissipation	Declared addendum	No change, γ 's fixed	Declared statistic $+0.038$ [$+0.013, +0.060$], a failure; swing (no rule) $+0.007$ [$-0.002, +0.019$]	Statistic responds to geometry; swing flat
Second mass ratio	Declared addendum; 18 of 20 seeds blind	Design-normalized $+0.070$	$+0.065$ [$+0.051, +0.081$]; blind $+0.058$ [$+0.047, +0.081$]	Supports the rise at a second point
Magnitude	Post hoc	Committed 0.165	Interventional 0.148 [0.143, 0.154] ($0.95\times$ forecast per observation; integrator check failed); background-subtracted 0.093; design-normalized nominal 0.161 [0.151, 0.167]	Approximate population magnitude, not a bound

comes instead from the recorded-geometry normalizer, which approaches or crosses zero at re-engagement, and from background motion of either sign inside the window, which can mask the snap response (8% of those observations record no drop). On them, the drop is 0.80 times the forecast at the recorded normalizer and 0.99 at the design normalizer. Applied post hoc, the design normalizer moves the cell medians to 0.132–0.232, putting two cells (0.208, 0.232) inside $[0.2, 0.9]$, and the nominal cell ($T_0 = 0.6$ kN, intensity 0.5) from 0.144 [0.127, 0.163] to 0.161 [0.151, 0.167]; these readings keep the background. For peaks below $2.3T_0$, the pre-registered ratio to the forecast lies within the declared factor 1.5 in all 15 evaluable cells (0.83–1.48, Fig. 4b); the fan cell at $T_0 = 1.0$ kN, intensity 0.5, has 12 such observations, below the declared minimum of 20 (unscored ratio 0.66).

A post hoc estimator, added after the pre-registered verdicts, removes the background: it re-integrates each snap from the recorded state 0.5 s before re-engagement under the recorded weather, once as recorded and once without the snapping cable's force, and takes the largest tension difference within one engagement period, on a 10 ms grid, per unit T_p (the interventional swing). It uses a separate 0.5 ms semi-implicit Euler model with constant thrust and headings replayed from the record, so the heading loop is not simulated. Its pre-registered check against the record failed on two criteria: other cables' slack onsets over 3 s were reproduced for 94.9% of 9646 re-engagements against 95% (61% in the $4k$ cell, at least 93.8% elsewhere), and one grazing contact re-engaged 5.2 ms late against a 5 ms limit; the median timing error is $6 \mu\text{s}$ and the 95th-percentile peak error 0.35%. The 3008 observations cover all 22 seeds of four cells and only the two pilot seeds of eleven others (624 from pilots; 2800 at $T_0 = 0.6$ kN). Their median swing is $0.148T_p$ [0.143, 0.154]; per observation it is 0.95 times the forecast, and cell medians range from 0.69 to 1.25

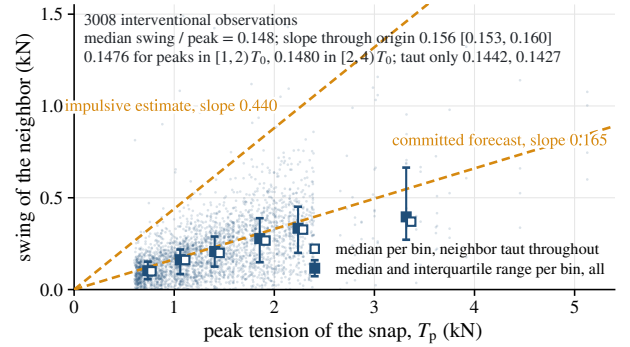


Fig. 5. Post hoc interventional swing versus snap peak for 3008 observations: medians and interquartile ranges per bin (filled) and medians for the 2766 observations whose neighbor stays taut (open). The through-origin slope is 0.156 [0.153, 0.160] (400 bootstrap replicates), versus 0.165 forecast and 0.440 impulsively.

times it. Subtracting each observation's own time-shifted drop instead (2928 observations with a background window) gives $0.093T_p$, a lower bound, since background and snap do not add inside a minimum; the gap between the two readings is the width of this measurement.

The interventional ratio is 0.1476 for $T_p \in [T_0, 2T_0]$ and 0.1480 for $T_p \in [2T_0, 4T_0]$ (Fig. 5), and 0.1442 and 0.1427 for neighbors that stay taut, each with a seed interval of ± 0.005 –0.008. They are consistent with, but do not prove, the amplitude invariance of Proposition 1.

C. Parameter Dependence and Unloading

Between $k/4$ and $4k$ the declared statistic rises from 0.125 to 0.161, $+0.036$ [$+0.015, +0.058$] over 20 paired seeds, against $+0.045$ predicted and zero from the undamped laws (Fig. 6a), the outcome declared as confirming. The median recorded geometry factor is flat in this pair (0.77 to 0.78), so the rise is in the drop itself (0.135 to 0.168 per unit peak). The interventional extremum occurs at 236, 139, and 80 ms

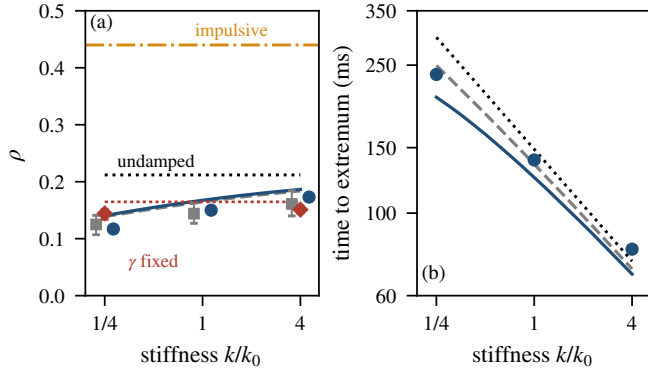


Fig. 6. Stiffness test in the nominal cell. (a) Declared statistic (squares, seed intervals) and post hoc interventional swing (circles) against the dissipative self-consistent (solid) and half-sine (dashed, the committed forecast) models, the undamped law (dotted), and the impulsive estimate (dash-dotted); diamonds: swing on the fixed-dissipation control cells against their flat forecast (red dotted); the control's declared statistic, 0.116 to 0.154, which fails that forecast, is not drawn. (b) Time from the snap to the swing's extremum against the same models.

for $k/4$, k , and $4k$ (Fig. 6b), a ratio of 2.96 against 2.99 for the self-consistent model, 3.5 for the half-sine pulse, and 4 undamped. That agreement is post hoc: the model times were computed after the measurements, the lags lie on a 10 ms grid, and the $k/4$ value rests on 13 snaps from the two pilot seeds.

A control declared before its runs scales c , c_L , c_A by $\sqrt{k/k_0}$ while varying k (nominal k_0), holding γ , γ_L , γ_A and the linearized forecast fixed at 0.165. By its declared rule it fails: the declared statistic rises by +0.038 [+0.013, +0.060]. The rise comes from the normalizer: as the scaled drag changes the shape the fleet holds at re-engagement, the share of observations with $g_{ij} \leq 0$ falls from 27 to 16% (median g_{ij} 0.69 to 0.85), while the drop per unit peak falls from 0.162 to 0.142. The interventional swing, declared as a second statistic without an outcome rule, carries neither background nor normalizer and is flat, 0.144 to 0.151, +0.007 [−0.002, +0.019], against 0.117 (13 pilot snaps) to 0.173 in the pre-registered pair; its lag ratio is 4.6 against 4.0 for the model. On the reading that isolates the transmission, stiffness enters through dissipation, as Proposition 1 states; the declared statistic is also sensitive to the fleet's geometry.

Pair structure. The post hoc swing improves the rank correlation with R_{ij} in the four fully re-integrated cells (0.76, 0.83, 0.95 in the parallel cells and 0.47 in the fan, against 0.24, 0.21, 0.58, 0.31 for the window drop on the same observations), but the spread of its pair medians within a cell is only 0.054–0.067, against 0.125–0.189 for R_{ij} .

A second mass ratio. A later addendum raised m_A from 600 to 1250 kg ($m_A/m_L = 0.50$), for which the collinear model predicts $0.167 \rightarrow 0.238$, printed before the run, and the planar linearization $0.165 \rightarrow 0.234$. It was written after a four-seed pipeline check whose two scored seeds were seen, and its primary statistic is the post hoc design-normalized reading. That statistic rises by +0.065 [+0.051, +0.081] against +0.070 predicted, and by +0.058 [+0.047, +0.081]

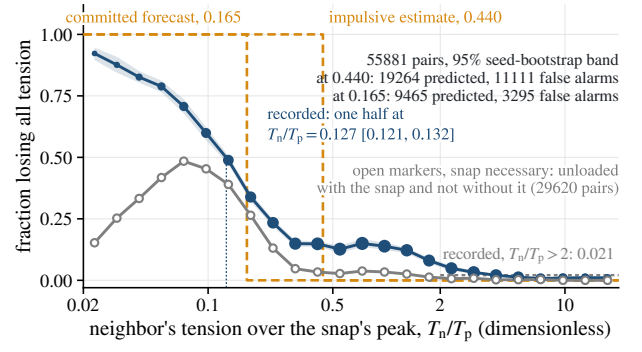


Fig. 7. Empirical analog of Corollary 1 over 55 881 (parent, taut neighbor) pairs. Each candidate ρ is a threshold on T_n/T_p (steps); the blue curve (marker area growing with pair count) is the recorded unloading probability with a 400-replicate seed band, crossing one-half at 0.127 [0.121, 0.132]. The gray curve (open markers), over 29 620 re-integrated pairs, is the fraction unloaded in the recorded branch but not the counterfactual one; the dotted level, 0.021, is the recorded rate at $T_n/T_p > 2$.

on the 18 unseen seeds. The declared secondary statistic, at the recorded geometry, rises by +0.046 [+0.022, +0.073] (0.144 $\rightarrow$ 0.189), and by +0.038 [+0.017, +0.068] on the unseen seeds; no rule was declared for it.

Unloading criterion. Fig. 7 scores T_n/T_p over 55 881 pairs of a declustered parent and a taut neighbor; 19.2% unload, and 45.8% of the 18 116 with $T_p > 4T_0$. It is an analog rather than a test of Corollary 1, which uses $\bar{T}_p$ where the figure uses the recorded T_p ; the two differ when a neighbor unloads early, as 41% of unloaded neighbors do within 60 ms, near or before the parent's peak. The unloading probability crosses one-half at 0.127 [0.121, 0.132]; as classifiers, $\rho_{\text{imp}} = 0.440$ recovers 8153 of 10 735 unloadings with 11 111 false alarms, and 0.165 recovers 6170 with 3295. The gray curve counts a neighbor whose elongation crosses zero within 3 s in the recorded branch but not in the counterfactual one, the snap as a necessary cause; since the integrator's pre-registered check failed, the declaration requires these counts to be reported as untrusted. It peaks near one-half for $T_n/T_p \in [0.05, 0.15]$ and is 0.004 for $T_n/T_p > 2$ (0.018 over all $T_n/T_p > 0.44$, where neither candidate predicts unloading). The planar transition is therefore population-level, not a deterministic bound on every pair.

VI. CONSEQUENCES, LIMITATIONS, AND CONCLUSION

On the nonlinear fleet, the collinear theory and the simulation agree on the magnitude and on its stiffness and mass dependence, but not on the distribution over pairs. The post hoc magnitudes, 0.148 by intervention and 0.161 by design normalization, lie within about 10% of the committed 0.165, above the background-subtracted bound 0.093 and far below the impulsive 0.44.

By Corollary 1, a snap unloads a neighbor at T_n once $\bar{T}_p$ reaches T_n/ρ : at pretension, $2.3T_0$ for the impulsive model, $6.1T_0$ and $6.8T_0$ for the committed and interventional magnitudes, and $10.8T_0$ for the background-subtracted bound. These are extrapolated scale estimates, not design bounds: the magnitude sample stops at $4T_0$, and zero tension censors

the taut response. Neighbors of scored snaps carry per-cell median tensions of $1.1\text{--}2.0T_0$, so $\rho \approx 0.15$ places isolated unloading near $7\text{--}14T_0$, several times the median parent peak of $2.5T_0$; yet one parent snap in five exceeds $7T_0$. Of the 10735 recorded unloadings in Fig. 7, 69% struck neighbors at or above pretension and 60% followed peaks above $7T_0$; the counterfactual counts, reported as untrusted, sharpen this to 79% and 82% of the 4336 unloadings for which the snap was a necessary cause (78% and 76% after reweighting for the re-integration sampling). Snap-driven unloading on this testbed thus comes mainly from the upper tail of snap peaks acting on normally loaded neighbors, not from neighbors near slack. The finite-contact law should not be read as bounding how readily slack spreads: the pre-registered test that turned R_{ij} and the neighbor margins into predicted slack onsets failed, under-predicting them in all 16 cells by factors of 0.39–0.80 (0.65–0.98 against a post hoc target of neighbors taut at the snap). Cable severance, which redistributes steady tension rather than transmitting a snap, is treated in [5].

The conclusions have three limits. The exact theory is collinear with lumped cables, whereas distributed lines carry the snap as a wave, and whether lever arms and yaw can raise a pair above the collinear worst case is open. The planar testbed has linear drag and no hull contact, and the interventional estimator (whose integrator failed its pre-registered check), the design normalization, and the timing comparison are post hoc. Two mass ratios, 0.24 and 0.50, test the rising branch of Fig. 3; the interior maximum near 1.4 is numerical, not measured or proved. Measuring it, and the pairwise distribution of ρ with its upper tail, on the planar fleet is future work.

Data, code, and pre-registration. The declaration record (SHA-256 prefix `fc26c18c14b97b04`) fixed the plan, cells, tests, pass criteria, and forecast before the campaign, pins the digests of the plan, forecast, theory document, modules, and inputs it uses, and was never edited. Of eight dated addenda, one records a reducer-crash fix, four record post hoc sensitivities, corrections, or reporting constraints that change no declared test, threshold, population, or rule, and three declare the fixed-dissipation control and the step-size check before their runs and the second mass ratio after a four-seed pipeline check. The declarations, addenda, run records, analysis code, and figure scripts are available from the authors.

REFERENCES

- [1] J. Esposito, M. Feemster, and E. Smith, “Cooperative manipulation on the water using a swarm of autonomous tugboats,” in *Proc. IEEE Int. Conf. Robotics and Automation (ICRA)*, 2008, pp. 1501–1506.
- [2] P. Cheng, J. Fink, S. Kim, and V. Kumar, “Cooperative towing with multiple robots,” in *Algorithmic Foundation of Robotics VIII*, ser. Springer Tracts in Advanced Robotics. Springer, 2009, pp. 101–116.
- [3] Z. Du, R. R. Negenborn, and V. Reppa, “Cooperative multi-agent control for autonomous ship towing under environmental disturbances,” *IEEE/CAA J. Autom. Sinica*, vol. 8, no. 8, pp. 1365–1379, Aug. 2021.
- [4] K. Sreenath and V. Kumar, “Dynamics, control and planning for cooperative manipulation of payloads suspended by cables from multiple quadrotor robots,” in *Robotics: Science and Systems IX*, Berlin, Germany, June 2013.
- [5] H. Hajieghrary and P. Schmitt, “Passive fault tolerance through tension-to-thrust feed-forward: Hybrid input-to-state stability for decentralized multi-UAV slung-load transport under abrupt cable severance,” arXiv:2605.05339, 2026. [Online]. Available: <https://arxiv.org/abs/2605.05339>
- [6] M. Carricato and J.-P. Merlet, “Stability analysis of underconstrained cable-driven parallel robots,” *IEEE Trans. Robot.*, vol. 29, no. 1, pp. 288–296, 2013.
- [7] J.-J. Moreau, “Unilateral contact and dry friction in finite freedom dynamics,” in *Nonsmooth Mechanics and Applications*, ser. CISM Courses and Lectures, J.-J. Moreau and P. D. Panagiotopoulos, Eds. Vienna: Springer, 1988, vol. 302, pp. 1–82.
- [8] B. Brogliato, *Nonsmooth Mechanics: Models, Dynamics and Control*, 3rd ed., ser. Communications and Control Engineering. Cham: Springer, 2016.
- [9] O. Khatib, “Inertial properties in robotic manipulation: An object-level framework,” *Int. J. Robot. Res.*, vol. 14, no. 1, pp. 19–36, 1995.
- [10] C. Liu, Z. Zhao, and B. Brogliato, “Frictionless multiple impacts in multibody systems. I. Theoretical framework,” *Proc. R. Soc. A*, vol. 464, no. 2100, pp. 3193–3211, 2008.
- [11] H. Wang, H. Li, B. Zhou, F. Gao, and S. Shen, “Impact-aware planning and control for aerial robots with suspended payloads,” *IEEE Trans. Robot.*, vol. 40, pp. 2478–2497, 2024.
- [12] J. M. Niedzwecki and S. K. Thampi, “Snap loading of marine cable systems,” *Appl. Ocean Res.*, vol. 13, no. 1, pp. 2–11, 1991.
- [13] S. Huang and D. Vassalos, “A numerical method for predicting snap loading of marine cables,” *Appl. Ocean Res.*, vol. 15, no. 4, pp. 235–242, 1993.
- [14] M. Hann, “Statics and dynamics of multi-cable systems for submersibles,” *Mar. Struct.*, vol. 8, no. 5, pp. 555–583, 1995.
- [15] C. M. Hennessey, N. J. Pearson, and R. H. Plaut, “Experimental snap loading of synthetic ropes,” *Shock Vib.*, vol. 12, no. 3, pp. 163–175, 2005.
- [16] W.-T. Hsu, K. P. Thiagarajan, and L. Manuel, “Extreme mooring tensions due to snap loads on a floating offshore wind turbine system,” *Mar. Struct.*, vol. 55, pp. 182–199, 2017.
- [17] M. Hann, D. Greaves, and A. Raby, “Snatch loading of a single taut moored floating wave energy converter due to focussed wave groups,” *Ocean Eng.*, vol. 96, pp. 258–271, 2015.
- [18] D. Qiao, J. Yan, H. Liang, D. Ning, B. Li, and J. Ou, “Analysis on snap load characteristics of mooring line in slack-taut process,” *Ocean Eng.*, vol. 196, p. 106807, 2020.
- [19] J. Palm, C. Eskilsson, and L. Bergdahl, “An hp-adaptive discontinuous Galerkin method for modelling snap loads in mooring cables,” *Ocean Eng.*, vol. 144, pp. 266–276, 2017.
- [20] N. Michael, J. Fink, and V. Kumar, “Cooperative manipulation and transportation with aerial robots,” in *Proc. Robotics: Science and Systems*, Seattle, WA, USA, June 2009.
- [21] M. Hall and A. Goupee, “Validation of a lumped-mass mooring line model with DeepCwind semisubmersible model test data,” *Ocean Eng.*, vol. 104, pp. 590–603, 2015.
- [22] K. H. Hunt and F. R. E. Crossley, “Coefficient of restitution interpreted as damping in vibroimpact,” *J. Appl. Mech.*, vol. 42, no. 2, pp. 440–445, 1975.
- [23] C. M. Harris and A. G. Piersol, Eds., *Harris’ Shock and Vibration Handbook*, 5th ed. New York: McGraw-Hill, 2002.
- [24] R. Tedrake and the Drake Development Team, “Drake: Model-based design and verification for robotics,” 2019. [Online]. Available: <https://drake.mit.edu>
- [25] B. A. Nosek, C. R. Ebersole, A. C. DeHaven, and D. T. Mellor, “The preregistration revolution,” *Proc. Natl. Acad. Sci. USA*, vol. 115, no. 11, pp. 2600–2606, 2018.
- [26] B. S. Siepe, F. Bartoš, T. P. Morris, A.-L. Boulesteix, D. W. Heck, and S. Pawel, “Simulation studies for methodological research in psychology: A standardized template for planning, preregistration, and reporting,” *Psychol. Methods*, 2024.